

2.5 Numerical techniques for solving the Anderson impurity model

We have seen that approximate analytical solutions of the AIM can be obtained via the equation of motion technique. Now we want to obtain **numerical solutions**.

For convenience we define the AIM:

$$H_{\text{AIM}} = H_{\text{atom}} + H_{\text{bath}} + H_{\text{hybridization}}$$

$$\begin{array}{ccc} \swarrow & \downarrow & \searrow \\ \underline{U n_{\uparrow\downarrow}^c + (\epsilon_0 - \mu)(n_{\uparrow}^c + n_{\downarrow}^c)} & \sum_{e\sigma} \tilde{\epsilon}_e a_{e\sigma}^\dagger a_{e\sigma} & \sum_{e\sigma} V e a_{e\sigma}^\dagger c_\sigma + V e^* a_{e\sigma} c_\sigma^\dagger \\ \text{interaction} & & \end{array}$$

In the EOM technique, we have seen an approximation for $G^R(\omega)$. Let us rewrite this in Matsubara frequencies:

$$G(i\omega_n) = \frac{1 - \langle n d \rangle}{i\omega_n - \epsilon_0 + \mu - \Sigma(i\omega_n)} + \frac{\langle n d \rangle}{i\omega_n - \epsilon_0 + \mu - \underline{U} - \Sigma(i\omega_n)}$$

with $\Sigma(i\omega_n) = \sum_e \frac{|V_e|^2}{i\omega_n - \tilde{\epsilon}_e}$

We now want to include the **interaction part** into the self-energy. However, at the same time, we want to separate the part which stems from the **hybridization**.

$$\Rightarrow \text{hybridization Green} \quad \Delta(i\omega_n) = \sum_e \frac{|V_e|^2}{i\omega_n - \tilde{\epsilon}_e}$$

(formerly "self-energy" Σ)

This leads to the definition of the non-interacting impurity Green function or Weiss field:

$$g_0(i\omega_n) = \frac{1}{i\omega_n + \mu - \epsilon_0 - \Delta(i\omega_n)}$$

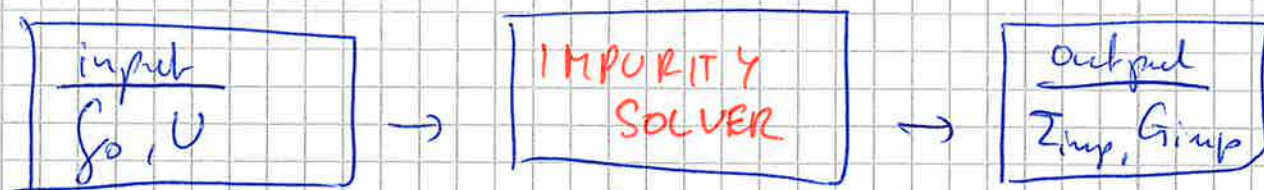
with which we can write the Dyson equation:

$$G_{\text{imp}}^{-1}(i\omega_n) = g_0^{-1}(i\omega_n) - \Sigma_{\text{imp}}(i\omega_n)$$

of the impurity problem.

Together with the interaction value U , the Weiss field g_0 determines the parameters of the AIM, since it includes the atomic energy level, chemical potential and both hoppings.

What we want to do now is:

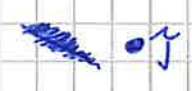


e.g., Hubbard-I,
IPT,
EO,
CTQMC,
...

We introduce this method here for the particle-hole (ph) Symmetric case, where the density of states $D(\omega) = D(-\omega)$ and the system is half-filled $\langle n_\uparrow + n_\downarrow \rangle = \langle n \rangle = n = 1$.

In the general case (out of ph symmetry) this method fails (due to a violation of the Luttinger theorem).

We draw Feynman diagrams for the self-energy (no external legs) and in $\gamma =$

→ non-iterative propagator $G_0(\gamma)$ 
 → iterative vertex

1st order $\Sigma^{1st}(\gamma) = \text{diagram} = (-1)^n (-1)^F U G_0(0^-) \delta_\gamma$
 $= \frac{U n}{2} \delta_\gamma = \frac{U}{2} \delta_\gamma$

$\Rightarrow \Sigma^{1st}(i\omega_n) = \frac{U}{2}$ Hartree term

For a spinful system with local moments there is no Fock term.

Second order $\Sigma^{2nd}(\gamma) = \text{diagram} =$

$= (-1)^n (-1)^F \overbrace{(-1)}^{-1} (G_0^2(\gamma)) G_0(-\gamma) = \left| \begin{matrix} \text{ph} \\ \text{symmetry} \\ G_0(-\gamma) = -G_0(\gamma) \end{matrix} \right| = +U^2 G_0^3(\gamma)$

$\Rightarrow \Sigma^{2nd}(i\omega_n) = U^2 \int_0^\beta d\gamma e^{i\omega_n \gamma} G_0^3(\gamma)$

$\Rightarrow \boxed{\Sigma^{IPT}(i\omega_n) = \frac{U}{2} + U^2 \int_0^\beta d\gamma e^{i\omega_n \gamma} G_0^3(\gamma)}$ 1st self-energy (only valid for ph-symmetry!)

advantages: Simple, quick, also possible on real-freq. axis, also applicable to itinerant coupling ($\Rightarrow$ see DMFT)

disadvantages: violates Luttinger theorem $\Rightarrow$ not correct out of ph-symmetry

2.5.3 Discretized Baths

These methods are not based on a perturbative expansion $\rightarrow$ represent the (infinite) degrees of freedom of the bath by a **finite number** of degrees of freedom.

Two important steps: ① discretization of bath
② Solution of finite Hamiltonian

① discretization of bath

$$\zeta_0^{-1}(i\omega_n) = i\omega_n + \mu - \epsilon_0 - \underbrace{\sum_{e=1}^{\infty} \frac{|V_e|^2}{i\omega_n - \tilde{\epsilon}_e}}_{\Delta(i\omega_n)}$$

For the next steps, it is better to work in real frequencies. The spectral theorem gives:

$$\Delta(\omega + i\eta) = \int_{-\infty}^{\infty} d\omega' \frac{\tilde{\Delta}(\omega')}{\omega - \omega' + i\eta} \quad \text{with } \tilde{\Delta}(\omega) = -\frac{1}{\pi} \text{Im} \Delta(\omega + i\eta)$$

$\rightarrow$ written (like DOS) as $\tilde{\Delta}(\omega) = \sum_{e=1}^{\infty} |V_e|^2 \delta(\omega - \tilde{\epsilon}_e)$

For ∞ degrees of freedom (d.o.f.) the $(V_e, \tilde{\epsilon}_e)$ are unique.

→ discretization $\tilde{\Delta}_N(\omega) = \sum_{e=1}^N |V_e|^2 \delta(\omega - \tilde{\epsilon}_e)$

This, conversely, is **not unique** anymore!

How can $(V_e, \tilde{\epsilon}_e)$ be determined, so that bath is filled in a good way?

We go back to Matsubara frequencies:

$$\left(\int_0^N (i\omega_n) \right)^{-1} = i\omega_n + \mu - \epsilon_0 - \sum_{e=1}^N \frac{|V_e|^2}{i\omega_n - \tilde{\epsilon}_e}$$

It turns out that the following distance measure works well:

$$d = \frac{1}{u_{\max}} \sum_{u=0}^{u_{\max}} u_n \left| \left(\int_0^N (i\omega_n) \right)^{-1} - \left(\int_0^N (i\omega_n) \right)^{-1} \right|,$$

where the sum runs over the first $u_{\max}$ Matsubara frequencies and u_n are weights.

② Solving the Hamiltonian

$$H = H_{\text{alen}} + \sum_{\sigma e=1}^N V_p (a_{er}^\dagger c_\sigma + c_\sigma^\dagger a_{er}) + \sum_{\sigma e=1}^N \tilde{\epsilon}_e a_{er}^\dagger a_{er}$$

→ Small bath: **exact diagonalization** (ED) of H possible

$O(N!7)$ ⇒ exact Lehmann representation

$$G(\text{imp}) = \frac{1}{z} \sum_{ij} \frac{|c(i)c(j)|^2}{i\omega_n + E_i - E_j} (e^{-\beta E_i} + e^{-\beta E_j})$$

→ Intermediate: Lanczos (Krylov-space technique)

⇒ only most important states

→ Large: Matrix product states (MPS)

2.54 Quantum Monte Carlo techniques

-) Hirsch-Fye quantum Monte Carlo (HF-QMC)

"auxiliary-field" quantum Monte Carlo (QMC)

originally formulated by J. Hirsch & R. Fye (1986)
for treating a small number of magnetic impurities
in metals $\rightarrow$ afterwards for general hybridization functions

The first step in HF-QMC is to **discretize the imaginary time interval** into a finite number of time slices:

L slices for $[0, \beta]$
each of length $\Delta\beta = \beta/L$

$$\Rightarrow \text{partition function } Z = \text{Tr} e^{-\beta H} = \text{Tr} e^{-\Delta\beta L H} = \\ = \text{Tr} e^{-\sum_{j=1}^L \Delta\beta H} = \text{Tr} \prod_{j=1}^L e^{-\Delta\beta H}$$

H is the Hamiltonian of the AIM with kinetic and interaction terms $H = H_0 + H_U$. This formula is exact, since $[H, H] = 0$, but one has to evaluate the exponential!
 $\rightarrow$ only possible if H is split, but $[H_0, H_U] \neq 0$

$\Rightarrow$ Baker-Hausdorff formula:

$$e^{-\Delta\beta (H_0 + H_U)} = e^{-\Delta\beta H_0} e^{-\Delta\beta H_U} + \mathcal{O}(\Delta\beta^2)$$

$\Rightarrow$ This introduces a systematic error of order $\Delta\beta^2$

$$\Rightarrow Z = \text{Tr} \prod_{\hat{S}=1}^L e^{-\Delta\beta H_0} e^{-\Delta\beta H_U + \sigma(\Delta\beta U)}$$

time discretization + Baker-Hausdorff = **Suzuki-Trotter decomposition**

$$H_U = U_{\uparrow\uparrow} u_{\downarrow} - \frac{U}{2} (u_{\uparrow} + u_{\downarrow}) \quad (\text{particle-hole symmetry cons})$$

→ **Hubbard-Shalunovich Transformation**

$$e^{-\Delta\beta U_{\uparrow\uparrow} u_{\downarrow} + \Delta\beta \frac{U}{2} (u_{\uparrow} + u_{\downarrow})} = \frac{1}{2} \sum_{S=\pm 1} e^{\lambda S(u_{\uparrow} - u_{\downarrow})}$$

with $\lambda \Rightarrow \cosh \lambda = e^{\Delta\beta U/2}$

proof: matrix elements on both sides in local basis

⇒ interaction commutator is quadratic!

BUT: we have to introduce a new **Ising-spin variable S**

We can now calculate the Green function at each time slice (without proof):

$$G = \frac{1}{Z} \sum_S p(S) G(S)$$

Ising spin configurations ↑ **probability density**

This can be efficiently done with **importance sampling** (Metropolis-Hastings algorithm)

advantages: exact if $U_S \rightarrow \infty$ and converged
disadvantages: fermionic sign problem!

in QMC, $p(s)$ can also be negative
for fermionic systems $\Rightarrow$ incompatible with
properties of a probability distribution function!

one way out: $\tilde{p}(s) = |p(s)|$

$$\langle \text{sgn} \rangle_{\tilde{p}} = \langle \text{sgn}(p(s)) \rangle_{\tilde{p}}$$

$$\rightarrow \frac{1}{\langle \text{sgn} \rangle_{\tilde{p}}} \langle G(s) \text{sgn}(p(s)) \rangle_{\tilde{p}}$$

$\langle \text{sgn} \rangle_{\tilde{p}} \approx 1 \Rightarrow$ no sign problem
(e.g. particle-hole symmetry,
half-filled systems)

$\langle \text{sgn} \rangle_{\tilde{p}} \approx 0 \Rightarrow$ statistical noise high,
bad sampling quality,
large sampling errors
(e.g.: low temperatures,
strong interactions,
large systems)

-) Continuous-time QMC

diagrammatic expansion of partition function
or interaction (CT-INT) or hybridization (CT-HYB)
up to high orders (~ 100)

$\rightarrow$ State-of-the-art (as of 2025)

disadvantage: fermionic sign problem

